# Design work for *St George's Field*.

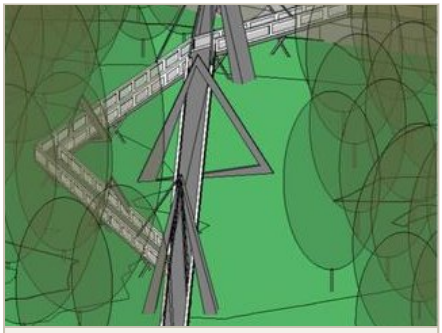
Three group design projects for one historic site on the University of Leeds campus: a cable-stayed raised walkway, a sheltering pavilion and a replacement boundary wall. Each one was taken from site analysis and precedent study through to Revit models, construction details, cost estimates and foundation calculations.

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| | |
|---|---|
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| TOOLS | Autodesk Revit, hand sketching, physical modelling |

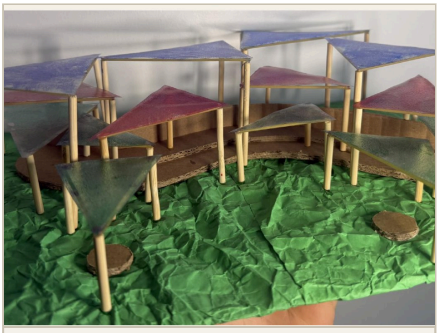
SELECTED PROJECTS — 2025/26



01 · BRIDGE / STRUCTURE

## Raised Walkway

A cable-stayed route across the field to first-floor level of the Sir William Henry Bragg Building.



02 · SHELTER / BUILDING

## Pavilion

45 steel columns and tinted glass panels arranged, in plan, as St George's dragon.




03 · WALL / FOUNDATION

## Vertical Barrier

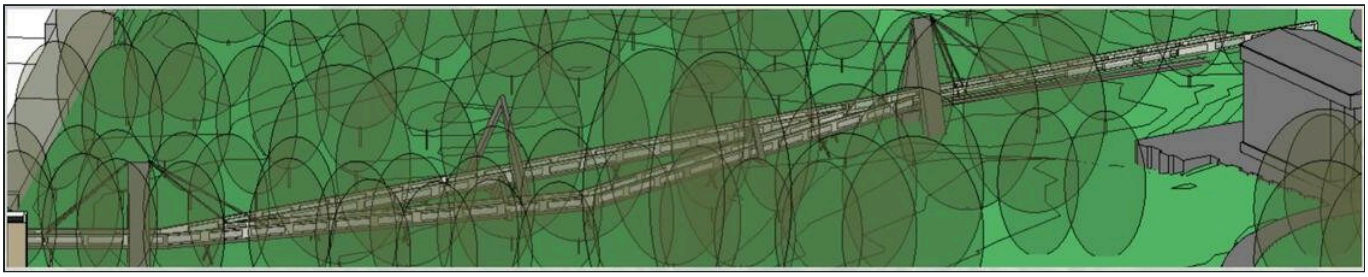
A new design to fill a 20 m gap in the field's stone boundary wall, with foundation calculations.

01

PROJECT 4 · GROUP PROJECT · JAN–FEB 2026

# St George's Field Raised Walkway

Cable-stayed | Composite deck | Weathering steel | Revit | Accessible design



LONG PERSPECTIVE: THE ROUTE RISING THROUGH THE TREE CANOPY TO THE BUILDING ABUTMENT

## THE BRIEF

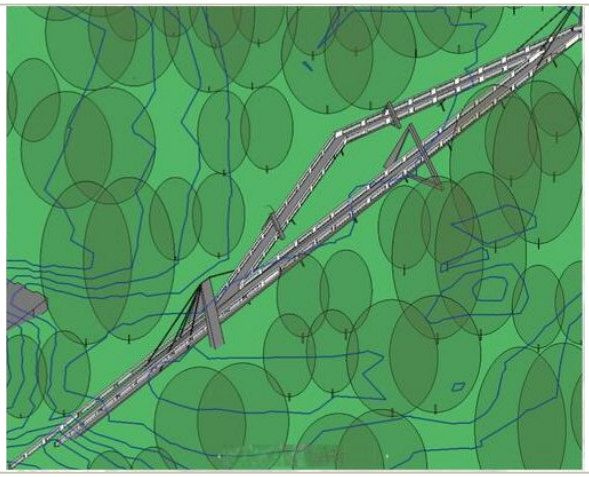
Design an accessible route across St George's Field, linking the existing path to first-floor level of the new Sir William Henry Bragg Building. The route has to rise over a protected masonry boundary wall without steps or steep ramps.

The constraints were a 2 m minimum clear width between handrails, no ground contact for 6 m from the building abutment, and an optional viewing platform no larger than 5 m².

## OUR RESPONSE

We designed a gently graded cable-stayed walkway. Weathering-steel pylons carry stay cables down to a composite steel–concrete deck. The supports are a series of hollow triangular reinforced-concrete frames that echo the pylons.

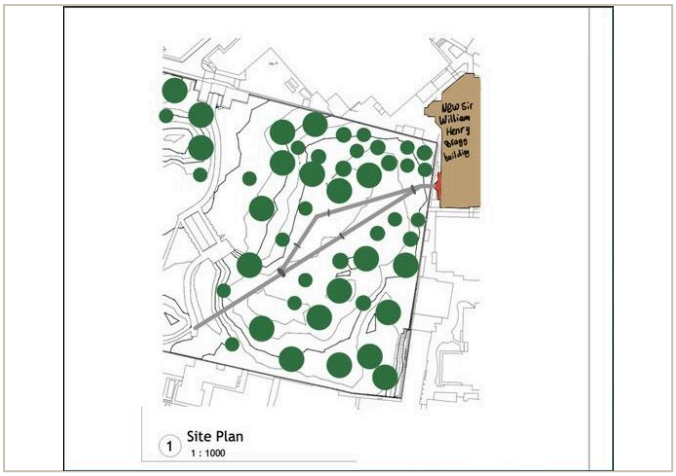
Laminated toughened glass balustrades keep the views open across the field. A slight cross-fall drains water to slot drains at the deck edges, which controls ice and protects the concrete from freeze–thaw damage.



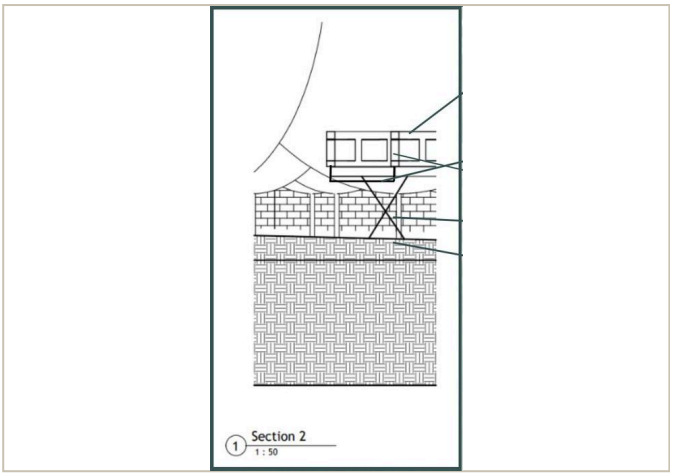
PLAN VIEW WITH TREE CANOPY AND CONTOURS




STRUCTURAL COMPONENTS: PYLON, DECK, SUPPORT




SITE PLAN, 1:1000




SECTION 2, 1:50: DECK, COLUMNS, FOUNDATIONS

## STRUCTURE & MATERIALS

- **Pylons:** weathering steel. It forms its own protective patina, so it needs no coating and less maintenance outdoors.
- **Deck:** steel beams with a concrete overlay, joined by welded shear studs so the two act as one composite section.
- **Supports:** hollow triangular RC frames. These save self-weight, can be inspected from inside and are bolted to the beams.
- **Balustrade:** laminated toughened glass on steel edge beams with mechanical clamps and a steel handrail.
- **Foundations:** concrete pads below each V-shaped steel column.

## CONSTRUCTION SEQUENCE

- Prepare the site and cast the foundations for the deck and pylons.
- Erect and brace the weathering-steel pylons and primary members.
- Lift in the deck sections and fix them to the main structure.
- Install and tension the steel stay cables, anchored at the pylons.
- Fit the glass balustrade panels and the steel handrail.

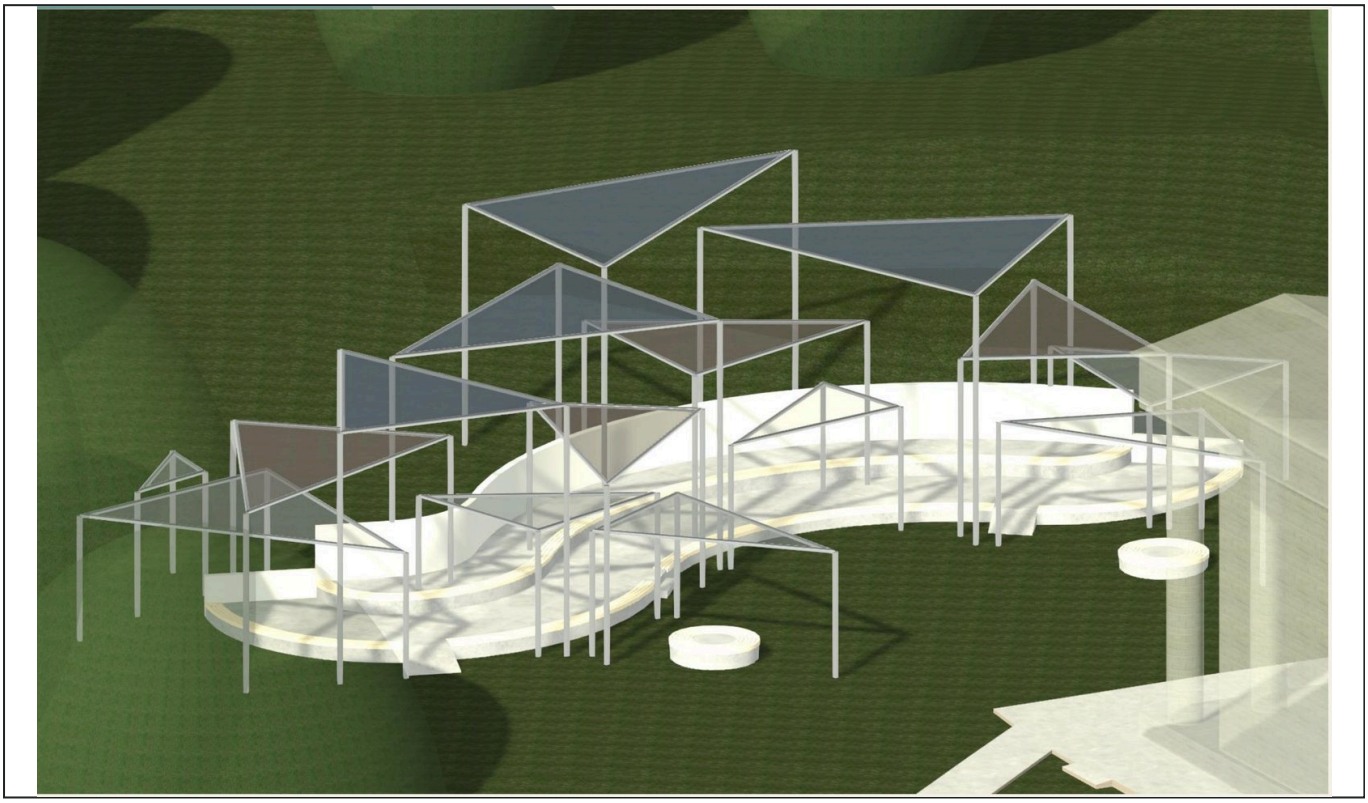
| 2.23 m | 500 mm | 3.5 m | 2 |
|---|---|---|---|
| DECK WIDTH | DECK DEPTH | GLASS PANEL WIDTH | CABLE-STAYED PYLONS |

02

PROJECT 5 · GROUP B3 · FEB–APR 2026

# St George's Field Pavilion

Steel frame | Glass canopy | Precast concrete | 1:100 model | Cost estimate



3D ORTHOGRAPHIC, 1:100 (REVIT)

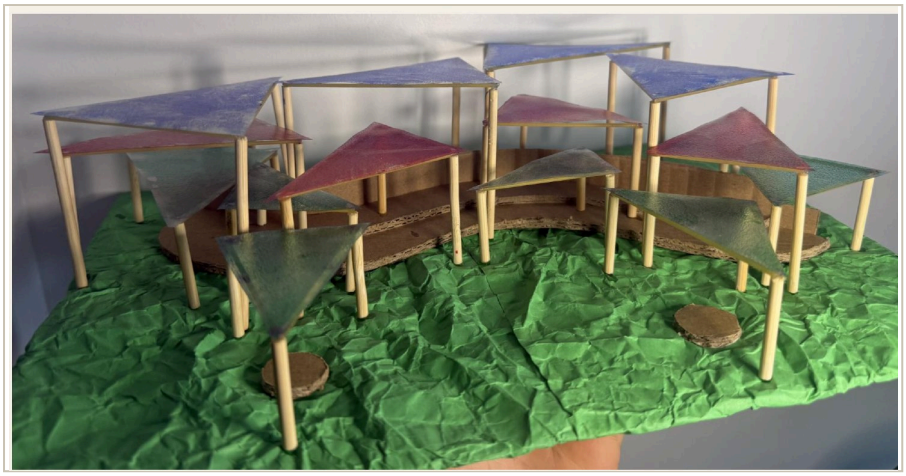
## THE BRIEF

Design an outdoor shelter on one general level that protects people from rain and gives some shelter from wind, without being fully enclosed. The covered floor area could be no more than 60 m², and any level changes had to use ramps rather than steps.

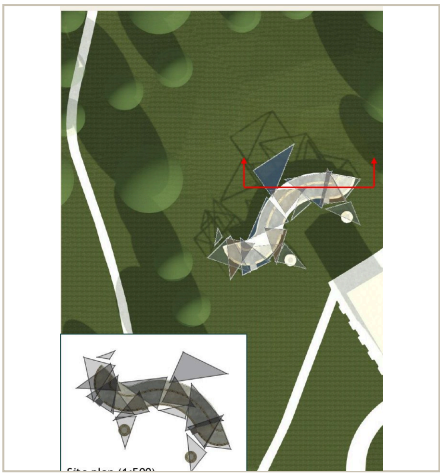
## DESIGN CONCEPT

The site is named after St George, so we drew on his legend. In plan, the triangular roof panels are positioned to form a dragon. Underneath, a curved two-level concrete base is shaped like a couch, making a place to relax or study that is sheltered but still open to the field.

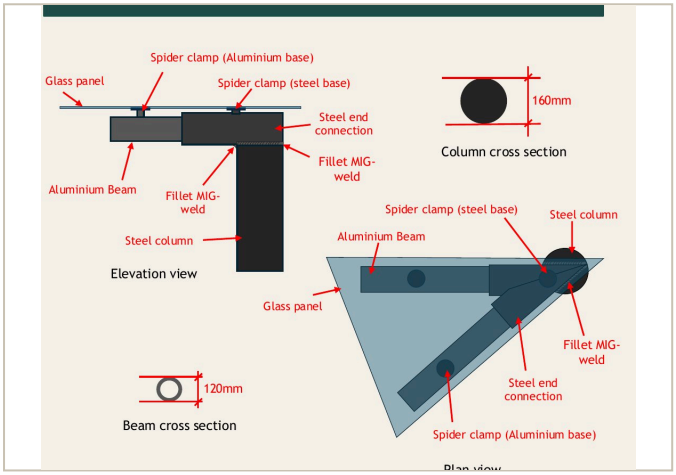
Tall roof panels, up to 7 m high, keep the space feeling open. A grid of 45 slender solid-steel columns gives a sense of safety and support.



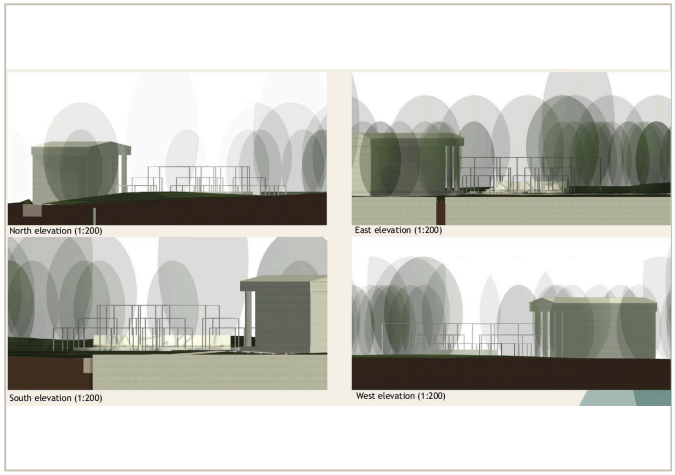
PHYSICAL MODEL, 1:100: BUILT FROM CARD, BAMBOO, SPAGHETTI AND PLASTIC WALLETS



SITE PLAN, 1:500




GLASS PANEL CONNECTION DETAIL, 1:5




ELEVATIONS, 1:200

## CONSTRUCTION METHOD

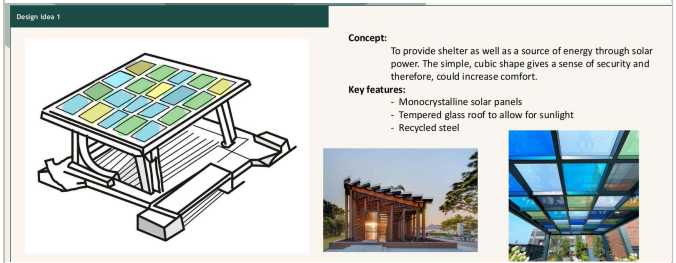
- Soil investigation, vegetation clearance and safe excavation, including bore holes for the column foundations.
- Pour reinforced foundations, then place the **precast** concrete base. Precasting cuts embodied carbon, and steel connectors join the blocks.
- Place the precast back wall, set the columns through the base into their bore holes, and fit varnished timber seating panels.
- Weld aluminium beams with steel ends to the columns, then fix the glass panels with spider clamps. All steel gets an anti-corrosion coating.

## KEY DETAILS

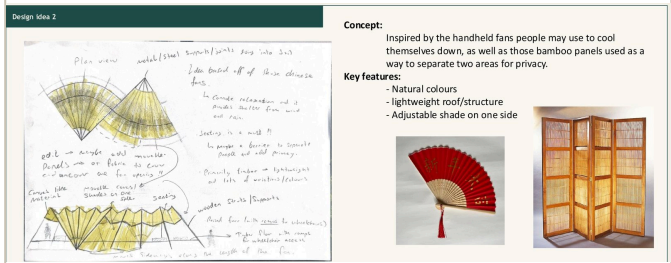
- **Canopy:** tinted glass panels that absorb UV radiation, keeping the space underneath cool.
- **Floor:** two concrete levels at 450 mm and 900 mm, with timber panels raising seat height to 480 mm. Ramps are provided for access.
- **Columns:** solid steel, embedded 2000 mm into the ground to resist buckling and overturning.
- **Form:** triangular roof panels, which are inherently stiff and strong.

## MATERIAL COST ESTIMATE

```
Reinforced concrete 464.1 m³ @ £200/m³ £92,817
Steel 4373.1 m³ @ £75/m³ £327,983
Aluminium 1.2 m³ @ £5,994/m³ £7,163
Spruce timber 7.7 m³ @ £800/m³ £6,139
Glass 334.4 m² @ £100/m² £3,344
TOTAL ESTIMATE £437,446
```




INITIAL IDEA 1: SOLAR-ROOF SHELTER




INITIAL IDEA 2: HAND-FAN CANOPY SKETCH

# 03

PROJECT 3 · GROUP PROJECT · NOV 2025–JAN 2026

# St George's Field Vertical Barrier

Masonry | Strip foundation | Wind loading | Overturning check | Revit section




FINAL DESIGN BOARD: ELEVATIONS, 3D VIEWS AND FOUNDATION CALCULATIONS

## THE BRIEF

Part of the field's stone boundary wall has collapsed, leaving a 20 m gap. Rather than copying the original, the replacement had to express current technology, culture and design thinking.

It had to be 2.5–4 m tall, fit within a 20 × 5 m zone and have no openings. Its external face could be no more than 10° off vertical. The ground dips sharply at mid-length: the soil can safely bear 50 kN/m² at 1 m depth, and rock at 4.5 m can bear 250 kN/m².

## OUR RESPONSE

Each team member developed an individual idea. We then combined the strongest parts of each into one final barrier, modelled it in Revit and developed it into a full presentation.

I produced an individual 2D Revit section through the wall and a foundation calculation. The calculation checks bearing pressure and overturning under the specified 1.2 kN/m² wind load.

## FOUNDATION CALCULATION SUMMARY

```
Wall height ..... 3.025 m
Strip foundation ..... 1.2 m wide × 0.4 m deep
Self-weight (wall + footing) ≈ 87.8 kN
Wind load ..... 1.2 kN/m² on the exposed face
Max bearing pressure ... ≈ 97 kN/m² vs allowable 250 kN/m² ✓ OK
Overturning: restoring moment ÷ overturning moment ≈ 4.5 > 1 ✓ STABLE
```

ABOUT

# Engineering that starts with the site.

I'm a civil engineering student at the University of Leeds. These projects come from CIVE1665 Integrated Design Project, a year-long design studio. In it, small teams take a real campus site from analysis and precedent research through to scaled drawings, Revit models, construction details, costing and structural checks.

I enjoy the point where an idea has to become buildable: choosing a material for how it weathers, working out how one element connects to the next, and checking a foundation can actually carry the load.

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## SOFTWARE

- Autodesk Revit
- Image editing for presentation boards
- Microsoft Office

## ENGINEERING

- Foundation design & bearing checks
- Wind loading & overturning
- Composite & cable-stayed structures
- Material selection & costing

## DESIGN

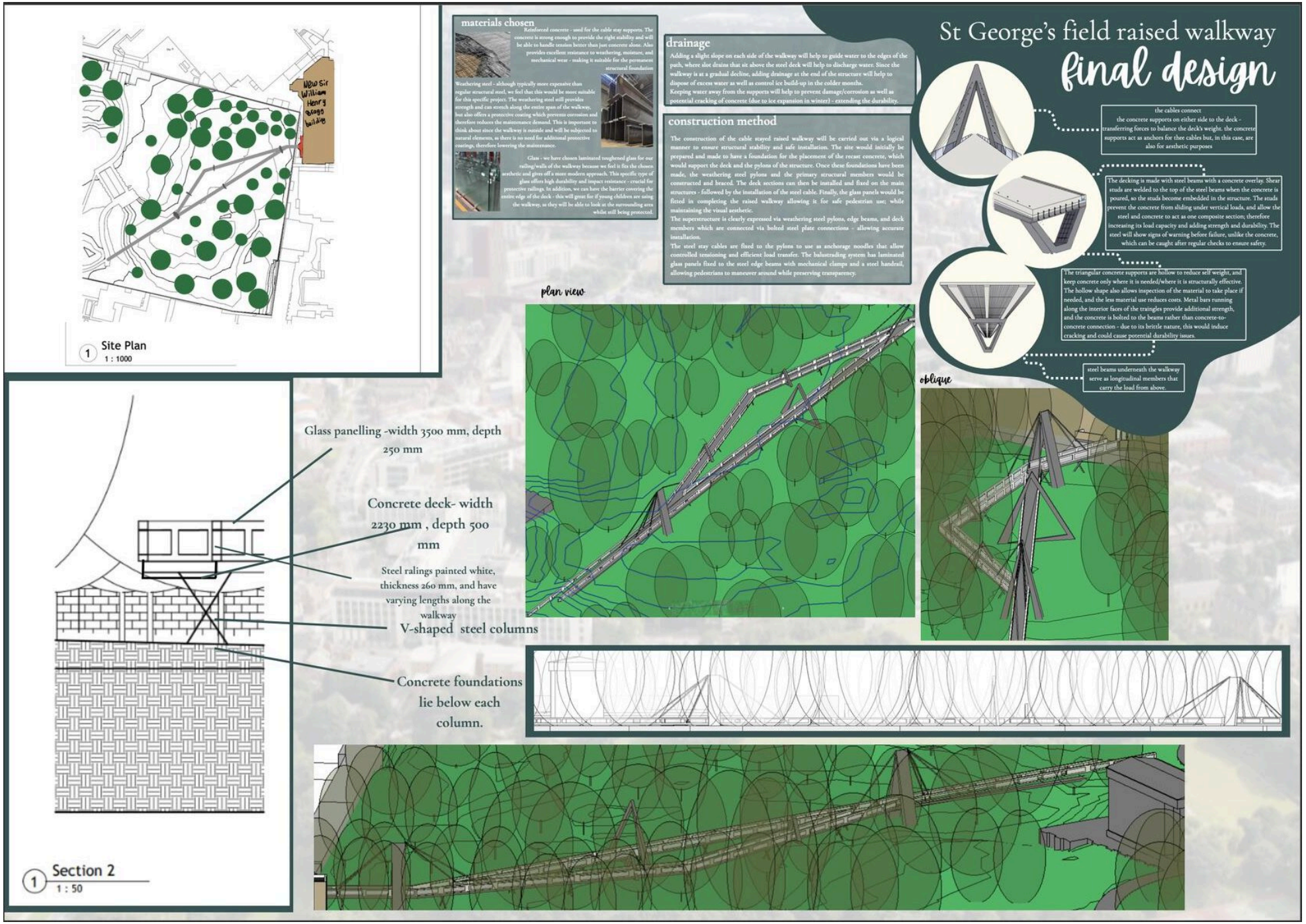
- Site analysis
- Precedent study
- Hand sketching
- Physical model making

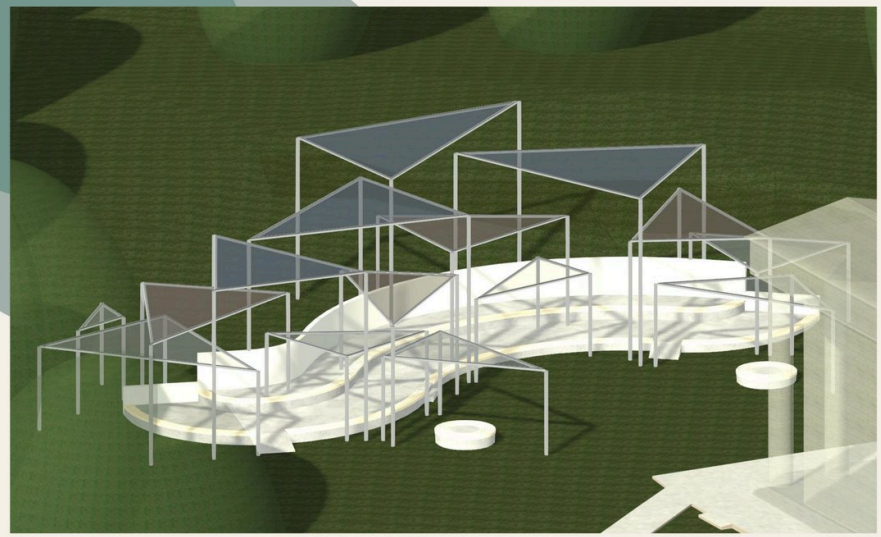
## DELIVERY

- A1 presentation boards
- Design reviews & crits
- Team collaboration
- Construction sequencing

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GROUP WORK PRODUCED WITH FELLOW STUDENTS, UNIVERSITY OF LEEDS SCHOOL OF CIVIL ENGINEERING

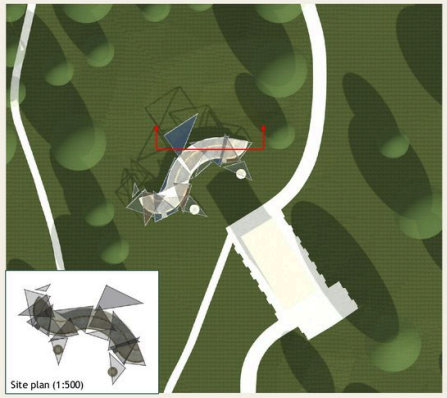

St George's field raised walkway
final design
materials chosen
Reinforced concrete - used for the cable stay supports. The concrete is strong enough to provide the right stability and will be able to handle tension better than just concrete alone. Also provides excellent resistance to weathering, moisture, and mechanical wear - making it suitable for the permanent structural foundation
Weathering steel - although typically more expensive than regular structural steel, we feel that this would be more suitable for this specific project. The weathering steel still provides strength and can stretch along the entire span of the walkway, but also offers a protective coating which prevents corrosion and therefore reduces the maintenance demand. This is important to think about since the walkway is outside and will be subjected to natural elements, as there is no need for additional protective coatings, therefore lowering the maintenance.
Glass - we have chosen laminated toughened glass for our railing/walls of the walkway because we feel it fits the chosen aesthetic and gives off a more modern approach. This specific type of glass offers high durability and impact resistance - crucial for protective railings. In addition, we can have the barrier covering the entire edge of the deck - this will great for if young children are using the walkway, as they will be able to look at the surrounding area whilst still being protected.
drainage
Adding a slight slope on each side of the walkway will help to guide water to the edges of the path, where slot drains that sit above the steel deck will help to discharge water. Since the walkway is at a gradual decline, adding drainage at the end of the structure will help to dispose of excess water as well as control ice build-up in the colder months.
Keeping water away from the supports will help to prevent damage/corrosion as well as potential cracking of concrete (due to ice expansion in winter) - extending the durability.
construction method
The construction of the cable stayed raised walkway will be carried out via a logical manner to ensure structural stability and safe installation. The site would initially be prepared and made to have a foundation for the placement of the precast concrete, which would support the deck and the pylons of the structure. Once these foundations have been made, the weathering steel pylons and the primary structural members would be constructed and braced. The deck sections can then be installed and fixed on the main structures - followed by the installation of the steel cable. Finally, the glass panels would be fitted in completing the raised walkway allowing it for safe pedestrian use; while maintaining the visual aesthetic.
The superstructure is clearly expressed via weathering steel pylons, edge beams, and deck members which are connected via bolted steel plate connections - allowing accurate installation.
The steel stay cables are fixed to the pylons to use as anchorage noodles that allow controlled tensioning and efficient load transfer. The balustrading system has laminated glass panels fixed to the steel edge beams with mechanical clamps and a steel handrail, allowing pedestrians to maneuver around while preserving transparency.
the cables connect the concrete supports on either side to the deck - transferring forces to balance the deck's weight. the concrete supports act as anchors for thee cables but, in this case, are also for aesthetic purposes
The decking is made with steel beams with a concrete overlay. Shear studs are welded to the top of the steel beams when the concrete is poured, so the studs become embedded in the structure. The studs prevent the concrete from sliding under vertical loads, and allow the steel and concrete to act as one composite section; therefore increasing its load capacity and adding strength and durability. The steel will show signs of warning before failure, unlike the concrete, which can be caught after regular checks to ensure safety.
The triangular concrete supports are hollow to reduce self weight, and keep concrete only where it is needed/where it is structurally effective. The hollow shape also allows inspection of the material to take place if needed, and the less material use reduces costs. Metal bars running along the interior faces of the traingles provide additional strength, and the concrete is bolted to the beams rather than concrete-to-concrete connection - due to its brittle nature, this would induce cracking and could cause potential durability issues.
steel beams underneath the walkway serve as longitudinal members that carry the load from above.
New Sir William Henry Stage building
1 Site Plan
1 : 1000
plan view
oblique
Glass panelling -width 3500 mm, depth 250 mm
Concrete deck- width 2230 mm , depth 500 mm
Steel ralings painted white, thickness 260 mm, and have varying lengths along the walkway
V-shaped steel columns
Concrete foundations lie below each column.
1 Section 2
1 : 50



3D Orthographic (1:100)

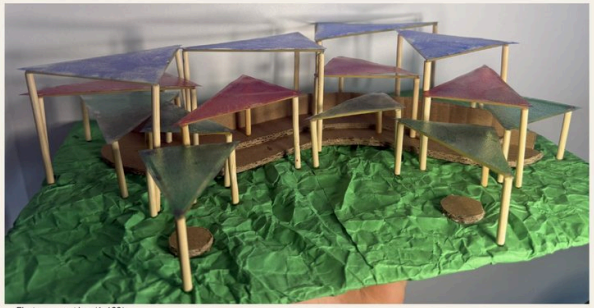
## Design concept

Our idea was to include context behind St George and his accomplishments while still making the pavilion functional as a place of relaxation/study sheltered from the elements but within them at the same time. The tall roof is there to make the space seem more open; and the shape of the base to imitate that of a couch. However, the 45 columns are there to provide a sense of safety and support, through their tall yet solid structure. When looked in plan view the panels' position and orientation resemble that of a dragon. This relates to the tale of how St George defeated a dragon to save a princess. The height of the roof panels further emphasizes this through their tall height (maximum 7000mm).

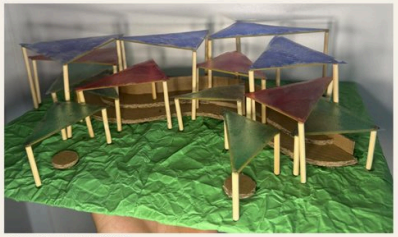


Site plan (1:500)

## Physical model (1:100 scale)



First perspective (1:100)

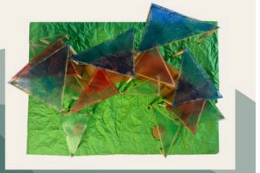


Second perspective (1:100)

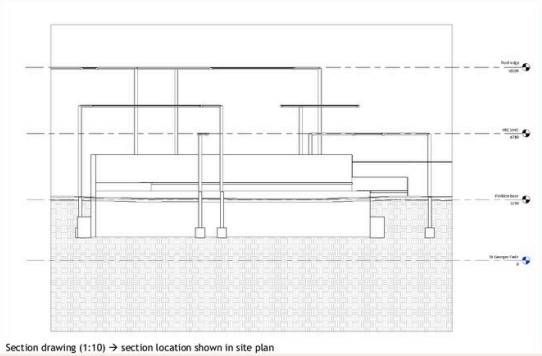
## Model information

The model was constructed using various resources:
Cardboard;
Coloured paper;
Bamboo sticks;
Spaghetti;
Plastic wallets;
Sharpie felt-tips;
Super-glue + activator and;
Clear tape.

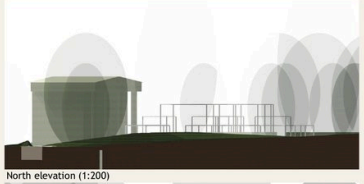
It is important to note that the model scale is only as estimate of 1:100 as it was challenging to measure and cut exact dimensions of the curved seating. Furthermore, the steel columns have a diameter of 180mm (as seen in the glass panel construction detail), this would mean that the 1:100 scale dimension for the model should be 1.8mm which is extremely thin, and therefore, the aluminium beams would be even thinner. This proved extremely difficult to work with so the scale for the columns and the beams are not accurate. However, the seating, and the size of the glass panels are to scale.



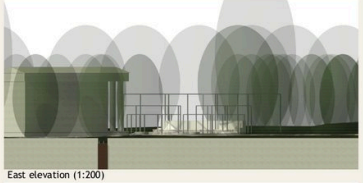
Plan view (1:200)



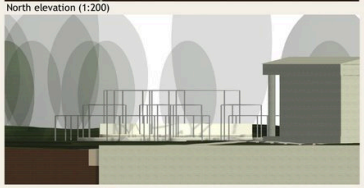
Section drawing (1:10) → section location shown in site plan



North elevation (1:200)



East elevation (1:200)



South elevation (1:200)

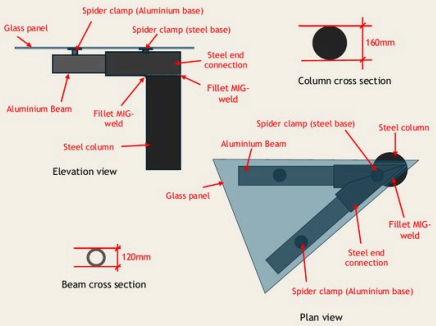


West elevation (1:200)

## Material cost estimation

| Material | Cost per unit … | Total area ($m^2$) | Total volume ($m^3$) | Total cost (£) |
|---|---|---|---|---|
| Reinforced Concrete | 200 £/$m^3$ | | 464.087 | 92817.4 |
| Steel | 75 £/$m^3$ | | 4373.1 | 327982.5 |
| Aluminium | 5994 £/$m^3$ | | 1.195 | 7162.83 |
| Spruce wood | 800 £/$m^3$ | | 7.673 | 6138.53 |
| Glass | 100 £/$m^2$ | 334.43 | | 3344.3 |
| **Total cost** | | | | **437445.56** |

## Glass panel construction detail (1:5)




## Canopy and enclosure

Glass panels are scattered over the pavilion in specific orientation and position as to for the shape of a dragon. The panels are tinted by a special material that absorbs UV radiation and therefore prevents magnification of most UV radiation, keeping the underside of the canopy cool.

## Floor

Features two levels, a lower, wider area and an upper, more narrow area; both constructed out of concrete. The floors are stacked and have been constructed with wooden panels that raise the seat height from 450mm to a more comfortable 480mm. Ramps are also constructed for those who may need them.

## Form and structure

In plan view, the shape of the pavilion takes the shape of a dragon (in context of the tale of St George), providing a sense of connectivity to St George and the site. The seating of the pavilion is relatively low, with its second level concrete base only standing at 900mm and its first at 450mm from the base of the structure. The slender columns may seem unstable and wea to buckling, however, they are constructed completely out of steel and are solid; they protrude 2000mm intro the surface from the base of the pavilion. The use of triangles as the roof panel shape also provides security through their inherited high-strength design.

## construction method

1. Firstly, the site must be prepared for excavation, this requires soil investigations and removal of plant/vegetation on the surface. Then the excavation must be dug (with appropriate shielding and safety procedures in place). Including excavating the holes for the columns and their foundations.

2. The foundation rebar and concrete must be placed and let to settle and dry. Then the construction of the base of the pavilion may be started. To reduce the embodies carbon, the base will be pre-cast and then transported to site; steel connections to be used to connect the blocks and filler to fill in the gaps.

3. After the fill has been set and all surfaces of the base grinded smooth, the back wall of the base will be placed (also pre-cast). Then the column foundations can be laid (as the pre-case bases have special placements for these that allow the columns that are situated on the base to be placed through the base and into the designated bore hole. Then the wooden seating panels can be attached to the base at the correct positions, and a clear varnish will be applied for extra protection from the elements.

4. The roof panels must then be prepared. Firstly, the columns will have thick aluminium beams connecting specific columns together. These will be solid and will have steel ends that will be welded onto the columns for a more seamless look. These beams will also have spider clamps tig-welded onto the aluminium beams which will be responsible for holding the glass panels in place. Then, all steel parts will be coated in an anti-corrosion applicant, preventing the steel columns and connections from corroding.

St George's Field

# Pavilion

*final design*

St George's Field

# Pavilion

*initial ideas*



Concept: An organic curved pavilions inspired by natural forms, creating a smooth flow from the ground up. The design forms a sheltered space, while the glass openings maintains a strong connection outdoors
Key Features: Smooth curved form- unique visual identity
Large glass openings.
Materials: Curved laminated timber-flexible, allows curved forms. Glass panels- brings in natural light and maintains connection to surroundings
Pros: Unique design and shape that resonates with surroundings
Cons: Not much privacy- glass openings make everything inside visible, Complex construction
Complexity level: Complex, curved structure requires skilled labour and project specific engineering



Material: Brick-Masonry-
Structurally durable
Key Features: Incision creating the entrance
Heavy grounded- blends into the ground space rather than differentiating itself from it
Semi-enclosed interior environment
Pros: Strong durable structural system
Natural material integration
Cons: Limited natural light, Project specific engineering
Concept: A continuous curved shell with a large vertical incision split in the middle forming the entrance, the design mimics a cave-like space embedded within the landscape



Materials: Panels- structural skin and wall. Glulam beams- Reinforces cantilever. Glass panels- allows light to enter
Key Features: Excessive cantilevered volumes
Sculptural but structural composition
Pros: Natural lighting
Cheap materials
Cons: Structurally complex
Needs precise engineering and detailing
Complexity level: Complex- project specific engineering, complex load paths, need to be very precise
Concept: Folded angular planes that create fragmented appearance, formed from intersecting triangular volumes projecting outwards

Concept: An organic pavilion inspired by natural cellular forms creating a sculptural and immersive space while providing a semi-enclosed space
Materials: Concrete (composite shell)- strong and is easily malleable. Perforated surface- tiny pores which allows air and light to get through.
Key Features: Perforations creates an interesting light pattern. Organic form blends in with natural concepts
Pros: Unique design, mechanically efficient- no need for glass panels for light due to perforation
Cons: Expensive to build



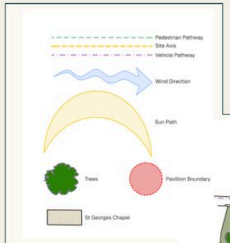
## Precedent research

| Material | Cost | Advantages | Disadvantages |
|---|---|---|---|
| Reinforced concrete | 150 – 250 £/m³ | Strong, durable, low maintenance | High embodied carbon, heavy |
| Glulam timber | 1000 £/m³ | Natural look, sustainable, warm | Requires maintenance, fire risk |
| Structural steel | 50 – 100 £/m³ | High strength, long life span | Corrosion risk, costly finish (for large projects) |
| Corten steel | 150 £/m³ | Self-weathering, unique aesthetic | Heavy, stains surroundings |
| Polyester canvas | 2.50 – 10 £/m² | Moisture and Mold resistant | Plastic look and feel, static electricity |
| glass | 35 – >100 £/m² | Seamless connection between interior and exterior | Greenhouse effect, poor insulation, can be fragile |





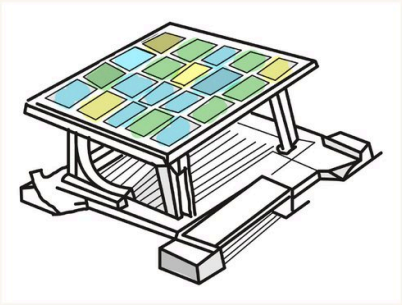
## Site Analysis






## Initial Design Ideas

### Design idea 1



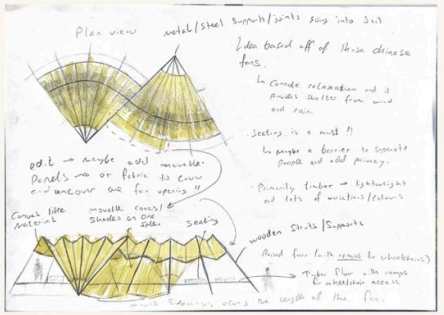
**Concept:**
To provide shelter as well as a source of energy through solar power. The simple, cubic shape gives a sense of security and therefore, could increase comfort.

**Key features:**
- Monocrystalline solar panels
- Tempered glass roof to allow for sunlight
- Recycled steel





### Design idea 2



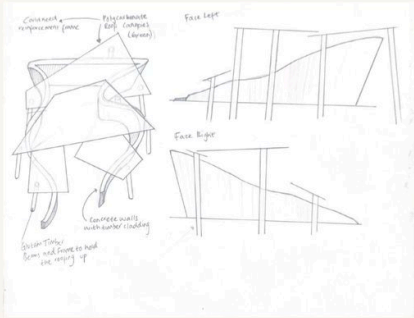

**Concept:**
Inspired by the handheld fans people may use to cool themselves down, as well as those bamboo panels used as a way to separate two areas for privacy.

**Key features:**
- Natural colours
- lightweight roof/structure
- Adjustable shade on one side





### Design idea 3



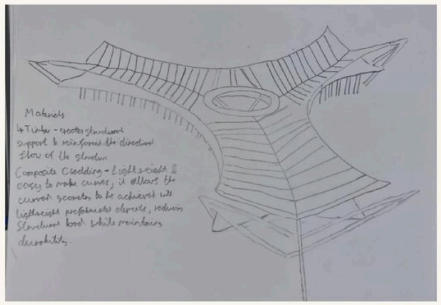

**Concept:**
Inspired by the shape of a high-heels with curved walls to mimic the fluidity of water

**Key features:**
- Poly-carbonate roof canopies
- Timber beams and frames
- Open walkway + space





### Design idea 4




**Concept:**
To provide a relatively large area of relaxation and comfort through the use of timber as the main material and cladding to provide a sense of shelter and cover.

**Key features:**
- Timber structure
- Composite cladding
- Large, curved roof

